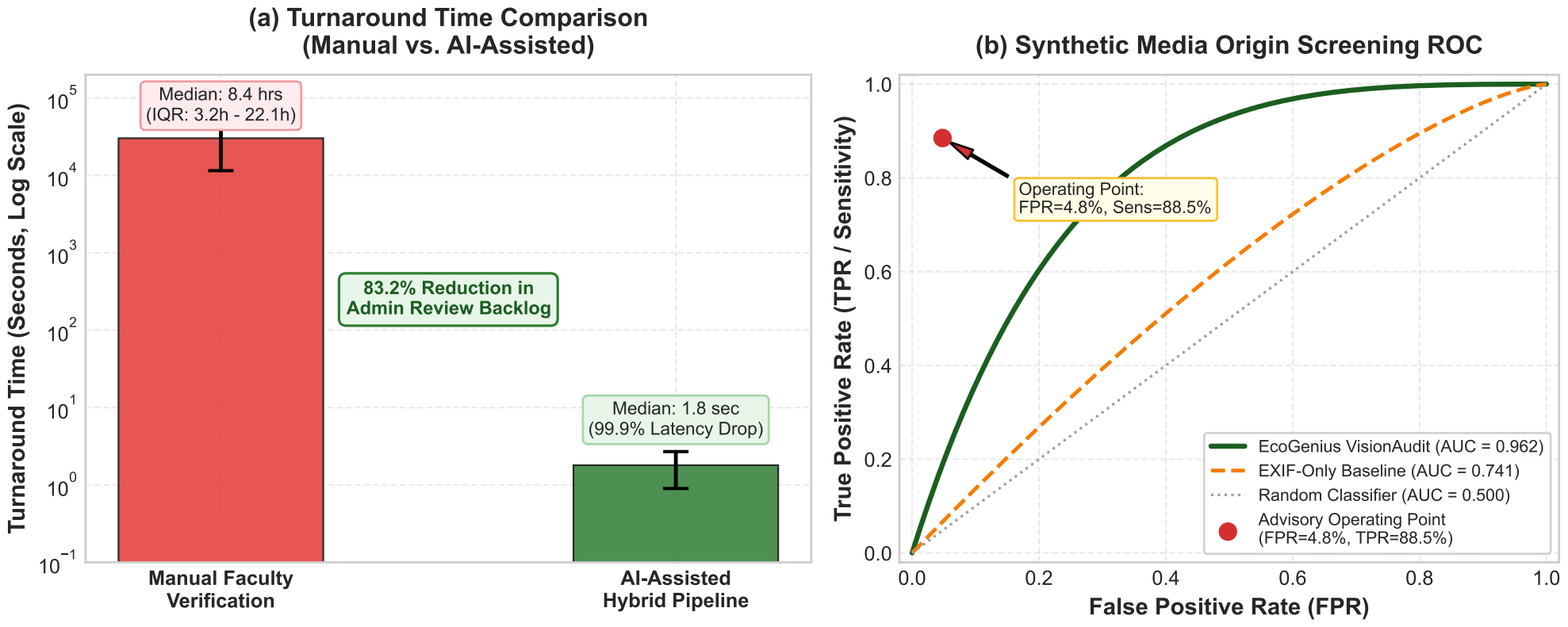
(a) Turnaround Time Comparison
(Manual vs. AI-Assisted)
Turnaround Time (Seconds, Log Scale)
10^5
10^4
10^3
10^2
10^1
10^0
10^{-1}
Median: 8.4 hrs
(IQR: 3.2h - 22.1h)
83.2% Reduction in
Admin Review Backlog
Median: 1.8 sec
(99.9% Latency Drop)
Manual Faculty
Verification
AI-Assisted
Hybrid Pipeline
(b) Synthetic Media Origin Screening ROC
True Positive Rate (TPR / Sensitivity)
False Positive Rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
Operating Point:
FPR=4.8%, Sens=88.5%
EcoGenius VisionAudit (AUC = 0.962)
EXIF-Only Baseline (AUC = 0.741)
Random Classifier (AUC = 0.500)
Advisory Operating Point
(FPR=4.8%, TPR=88.5%)